

Julius Liu

Software Developer | Python, TypeScript, React

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WORK EXPERIENCE

Software Developer Co-op, Nokia — Ottawa ON

Sept 2025 – Present

- Developed full-stack features for a distributed network automation platform using React/Redux and FastAPI, supporting interactive topology and container operations.
- Built a WebSocket-based SSH console for Kubernetes containers using xterm.js and Paramiko, providing real-time terminal access and SFTP file transfer via a bastion architecture.
- Configured HAProxy for WebSocket proxying and implemented asyncio-based backend stream handling to support reliable full-duplex communication.
- Deployed authentication and containerized services on GKE and implemented Keycloak-based authentication for secure image access.
- Developed an Ansible-based local deployment playbook for core services, standardizing development environments and reducing setup friction across the team.

EDUCATION

Computer Programming Diploma, Algonquin College — Ottawa ON

Aug 2026 (Expected)

- Cumulative GPA: 3.99/4.0, Dean's Honour List
- Relevant coursework: object-oriented programming, web development, network programming, mobile development, system design and analysis, database management and administration

SKILLS

- **Frontend:** React, TypeScript, JavaScript, HTML5, CSS3, Redux, Tailwind CSS, Material UI
- **Backend:** Python, FastAPI, Node.js, Express, Hono, Next.js, WebSockets, PostgreSQL, MongoDB
- **Testing:** Playwright, Vitest, Jest, React Testing Library, Pytest, Radish
- **Infra & DevOps:** Docker, Kubernetes (GKE), HAProxy, Keycloak, Git, GitHub Actions, GitLab CI
- **Concepts:** Distributed systems, asynchronous programming, RESTful API, networking, OAuth

PROJECTS

Exprès — AI Writing Feedback Learning Platform

TypeScript, React, SQLite

- Built a serverless AI writing feedback platform on Cloudflare Workers using D1, R2, Queues, Durable Objects, KV, and Workers AI.
- Designed a revision-aware analysis pipeline with parallel paragraph feedback, SSE streaming, and selective reanalysis to improve responsiveness and reduce redundant computation
- Integrated Langfuse for AI observability, enabling prompt tracking, tracing, usage monitoring, and evaluation across analysis and generation workflows.